

MD HASAN ANOWAR

Ames, IA 50010

956-360-1171 | hasan92niloy@gmail.com | www.linkedin.com/in/hasananowar |
<https://github.com/hasananowar> | <https://scholar.google.com/citations?user=uxVbaYYAAAAJ&hl=en>

PROFESSIONAL SUMMARY

3rd year Ph.D. student in Computer Engineering with 4 years of experience in data mining and machine learning. Research focuses on multidisciplinary research in spatial and temporal analysis of biomolecular simulation data.

EDUCATION

Iowa State University (ISU), Ames, IA Jan 2021-Present

Doctor of Philosophy, Ph.D., Computer Engineering

GPA : 3.92/4.00

Supervisor: Dr. Goce Trajcevski, Dr. Ashfaq Khokhar

The University of Texas Rio Grande Valley (UTRGV), Edinburg, TX Jan 2019-Dec 2020

Master of Science, M.S., Electrical Engineering

GPA : 4.00/4.00

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh May 2010-Sept 2015

Bachelor of Science, B.Sc., Electrical and Electrical Engineering

TECHNICAL and LABORATORY SKILLS

Programming: Python, MATLAB, C++, Java

Tools & Libraries: Numpy, Pandas, Scikit-learn, PyTorch, OpenCV, Matplotlib

Database: MDX, SQL, Cypher Neo4j, MongoDB

Big Data: Apache Spark, Hadoop

Simulation: Cadence, Simulink

Molecular Chemistry Tools: Amber, VMD

Miscellaneous: LaTeX, GitHub, Slack

WORK EXPERIENCE

Graduate Research Assistant, Trajectory Data Mining Jan 2021-Present

ECE Department, ISU, Ames, IA

- Analyze large molecular dynamics simulation trajectory data sets (>2,000 Gigabytes) using data mining algorithms.
- Conduct computational research using medical drug data to understand their behavior which impacts drug development technology.
- Developed a compression technique for the molecular trajectories participating in a chemical interaction.

Senior Design Project Mentor, Detecting Convoys of Molecular (and Atomic) Trajectories Aug 2022-Present

ECE Department, ISU, Ames, IA

- Co-mentoring a group of 6 students to develop a User-Interface tool that will allow chemists and researchers to visualize the trajectories of chemical objects.

Research Assistant, Brain-computer Interface Project Jan 2019-Dec 2019

EE Department, UTRGV, Edinburg, TX

- Developed a brain signal analysis toolbox to analyze and interpret biomedical signals
- Implemented supervised machine learning models for motor imagery movement detection and automatic detection algorithms for sleep feature identification.
- Studied EEG, and ECoG signals to extract biological information

Teaching Assistant, Digital Systems lab Electrical Circuits Department of EE, UTRGV, Edinburg, TX • Assist students in solving Combinational and sequential logic design problems • Taught laboratory class sessions containing 25+ students in each session • Graded homework and prepared lectures and instructional videos for the online classes	Jan 2020-Dec 2020
Assistant Engineer Bangladesh Rural Electrification Board, Dhaka, Bangladesh • Prepared standard operating procedures and project reports • Worked as an electrical engineer for the largest electrical power distribution board	Oct 2018-Dec 2018
Lecturer, Energy Conversion Control Systems Department of EE, City University, Dhaka, Bangladesh • Taught and designed syllabi of energy conversion I-II courses • Guide students to design PID controlled robots as part of laboratory projects	Dec 2015-Sept 2018
SELECT PUBLICATIONS	
• Anowar, M.H. et al. (2022). Generalization Aware Compression of Molecular Trajectories. In: Chiusano, S., Cerquitelli, T., Wrembel, R. (eds) Advances in Databases and Information Systems. ADBIS 2022. Lecture Notes in Computer Science, vol 13389. Springer, Cham. • Anowar, M.H. and Roy, P., "A Modified Incremental Conductance Based Photovoltaic MPPT Charge Controller" in IEEE International Conference on ECCE, Cox'sBazar, Bangladesh, Feb 2019 • Anowar, M.H. and Choi, Paul Y., "An Open Source Python App for Sleep Data Analysis" in Bioelectronics, Biosensors, and Biochips Virtual Meeting, May'2020 (Abstract Presentation)	
SELECT HONORS AND AWARDS	
• Best Paper Award for the paper "Generalization Aware Compression of Molecular Trajectories", in ADBIS 2022, Turin, Italy	Sept 2022
• Graduate student recognition, ISU ECE honors, and awards reception, Ames, IA	Oct 2022
• Finalist Award, Graduate Research Showcase, UTRGV, Edinburg, TX	Nov 2020
LEADERSHIP SKILLS AND SERVICE	
Treasurer, Graduate Organization of Electrical and Computer Engineering, ISU • Assist fellow graduate students in their professional development via social and technical activities. • Document and monitor the financial activity of the club.	Sept 2022-Present
Cultural Ambassador, ISU, Ames, IA • Communicated about variations of culture, rituals, work ethics, etc., with fellow ISU students • Made connections to the local and international student community through diverse and inclusive activities	Aug 2021-Dec 2021
Student Representative, Research and Educational Technology Committee Student Government Association, UTRGV, Edinburg, TX • Served as a liaison between the student body to the university governance • Worked to improve the methods and tools for quality education	Oct 2019-Dec 2020
Educator and Executive Committee Member Pothoshishu Sheba Songothon (http://www.pothoshishusheba.org/), Dhaka, Bangladesh • Provided education, medicine, and shelter to homeless children as a community volunteer • Trained to provide Therapeutic play skills, Non-formal teaching, and First aid.	2010-2018